Quick Introduction to Deep Learning

GENE 46100 — Unit 0

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2025-03-25

Course Roadmap

Unit	Topic
0	Can we detect regulatory motifs in DNA?
1–3)	
1	Can we learn the "language" of DNA?
2	Can we predict gene regulation from sequence?
3	Can we model microbial communities?

Tools we'll use:

- Python + PyTorch
- Google Colab (GPU)
- Weights & Biases

By the end: You'll understand how state-of-the-art models predict gene expression from DNA sequence

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What Can Deep Learning Do in Genomics?

- **Predict TF binding** from DNA sequence alone
- **Score regulatory variants** without experiments
- **Predict gene expression** from 200kb of sequence (Enformer)
- **Generate protein structures** (AlphaFold)
- **Build DNA language models** that learn grammar of the genome

. . .

These models learn patterns we didn't know to look for.

Today's Learning Objectives

1. Understand and implement **gradient descent** to fit a linear model
2. See why linear models **fail** on nonlinear data
3. Build a **multi-layer perceptron (MLP)** that succeeds
4. Learn the basics of **PyTorch**: model, loss, optimizer

The Starting Point: Linear Models

You already know this:

$$y = X\beta + \epsilon$$

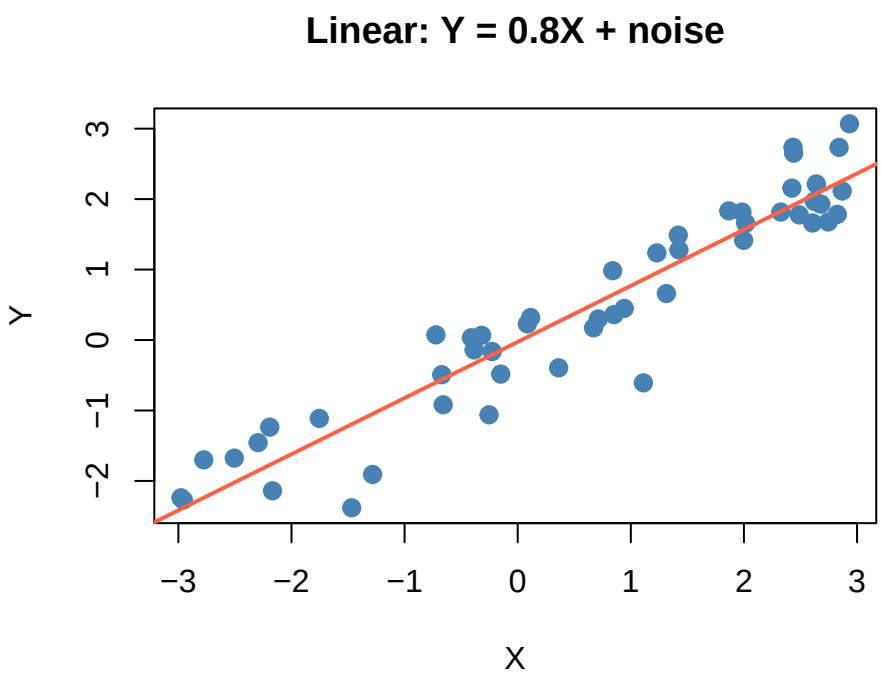
where $\epsilon \sim N(0, \sigma)$

. . .

Let's simulate some data and fit it **from scratch**.

- X: 1000 samples $\times$ 2 features
- y: response variable
- β: true coefficients

```
set.seed(42)
x <- runif(50, -3, 3)
y <- 0.8 * x + rnorm(50, sd = 0.5)
plot(x, y, pch = 19, col = "steelblue", cex = 1.2,
     xlab = "X", ylab = "Y", main = "Linear: Y = 0.8X + noise")
abline(lm(y ~ x), col = "tomato", lwd = 2)
```



Predict with Ground Truth Parameters

Since we simulated the data, we **know** the true β.

We can compute $\hat{y} = X\beta$ and compare to y:

```
y_hat = x.dot(coef) + bias
```

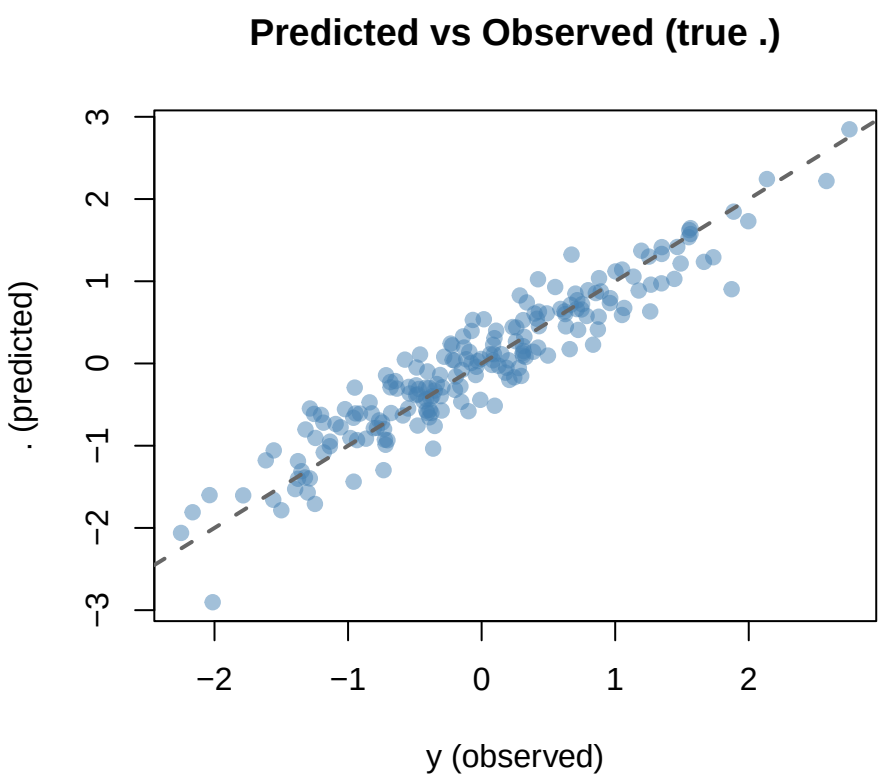
. . .

If the model is correct, points fall on the identity line.

The scatter off the line = noise ϵ.

```
set.seed(42)
n <- 200
x1 <- rnorm(n); x2 <- rnorm(n)
coef <- c(0.8, 0.5)
y <- x1 * coef[1] + x2 * coef[2] + rnorm(n, sd = 0.3)
y_hat <- x1 * coef[1] + x2 * coef[2]
plot(y, y_hat, pch = 19, col = adjustcolor("steelblue", 0.5), cex = 1,
```

```
     xlab = "y (observed)", ylab = "ŷ (predicted)",
     main = "Predicted vs Observed (true  )")
abline(0, 1, col = "gray40", lwd = 2, lty = 2)
```



The Analytical Solution: Normal Equation

For linear regression, there's a **closed-form** solution:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

. . .

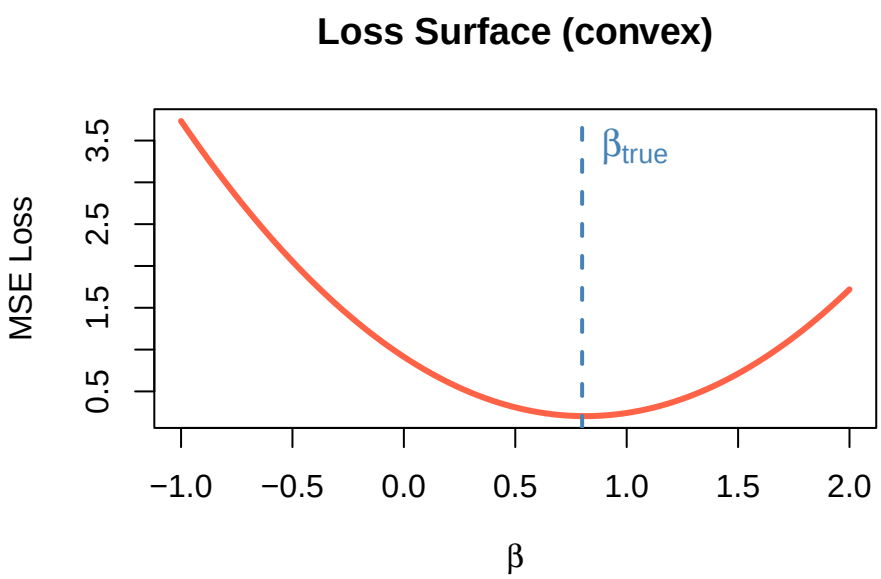
```
b_hat = inv(x.T @ x) @ (x.T @ y)
```

. . .

Estimated: $\hat{\beta} \approx$ true β

This works because linear regression has a **convex** loss surface — one global minimum.

```
# Loss surface for 1D regression
beta_seq <- seq(-1, 2, length.out = 100)
set.seed(42)
x_sim <- rnorm(100)
y_sim <- 0.8 * x_sim + rnorm(100, sd = 0.5)
loss <- sapply(beta_seq, function(b) mean((y_sim - b * x_sim)^2))
plot(beta_seq, loss, type = "l", lwd = 3, col = "tomato",
     xlab = expression(beta), ylab = "MSE Loss",
     main = "Loss Surface (convex)")
abline(v = 0.8, col = "steelblue", lwd = 2, lty = 2)
text(0.8, max(loss)*0.9, expression(beta["true"]), col = "steelblue", pos = 4, cex = 1.2)
```



But What if We Can't Solve It Analytically?

The normal equation works for linear models…

. . .

But most interesting models don't have closed-form solutions.

- Neural networks with millions of parameters?

- Nonlinear activation functions?
- Complex architectures?

. . .

We need a general-purpose optimization method.

What is a Gradient?

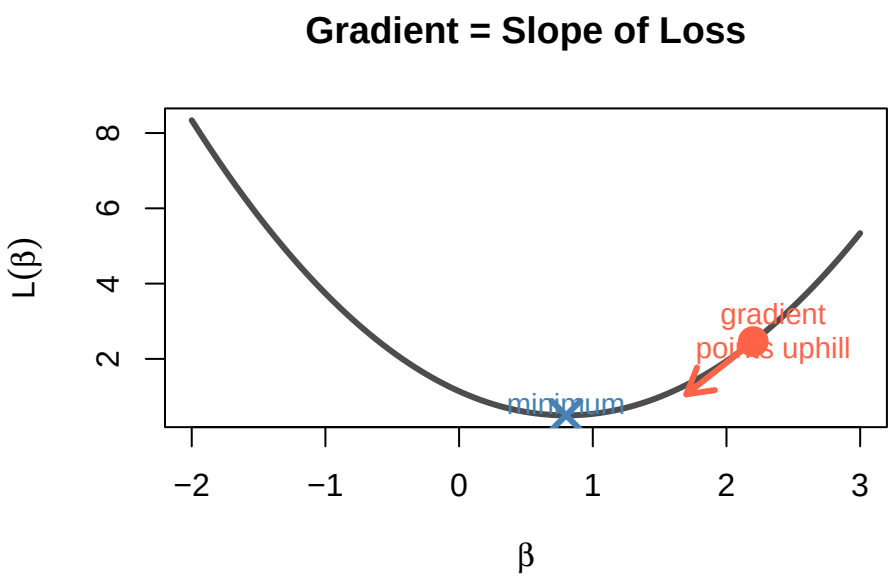
A gradient is the **derivative** of a function — it tells us the slope.

$$f'(\beta) = \lim_{\Delta\beta \to 0} \frac{f(\beta + \Delta\beta) - f(\beta)}{\Delta\beta}$$

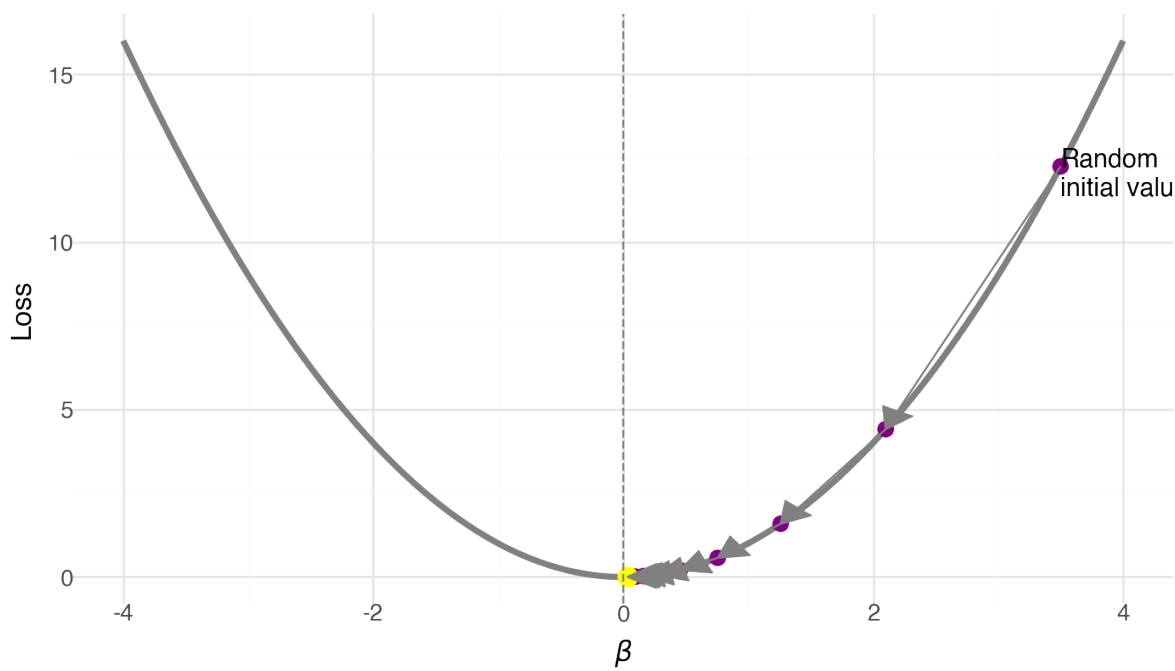
. . .

- **Positive gradient** → function is increasing → move left
- **Negative gradient** → function is decreasing → move right
- **Zero gradient** → you're at a minimum (or maximum)

```r
x_vals <- seq(-2, 3, length.out = 200)
y_vals <- (x_vals - 0.8)^2 + 0.5
plot(x_vals, y_vals, type = "l", lwd = 3, col = "gray30",
     xlab = expression(beta), ylab = expression(L(beta)),
     main = "Gradient = Slope of Loss")
# Show gradient at a point
pt <- 2.2
slope <- 2 * (pt - 0.8)
y_pt <- (pt - 0.8)^2 + 0.5
arrows(pt, y_pt, pt - 0.5, y_pt - 0.5 * slope, col = "tomato", lwd = 3, length = 0.15)
points(pt, y_pt, pch = 19, col = "tomato", cex = 2)
text(pt + 0.15, y_pt + 0.3, "gradient\npoints uphill", col = "tomato", cex = 0.9)
points(0.8, 0.5, pch = 4, col = "steelblue", cex = 2, lwd = 3)
text(0.8, 0.8, "minimum", col = "steelblue", cex = 0.9)
```



Gradient Descent: Follow the Slope Downhill



Algorithm:

1. Start at a **random** β (high loss)
2. Compute the **gradient** (slope)
3. Take a step **opposite** to the gradient
4. Repeat until convergence

. . .

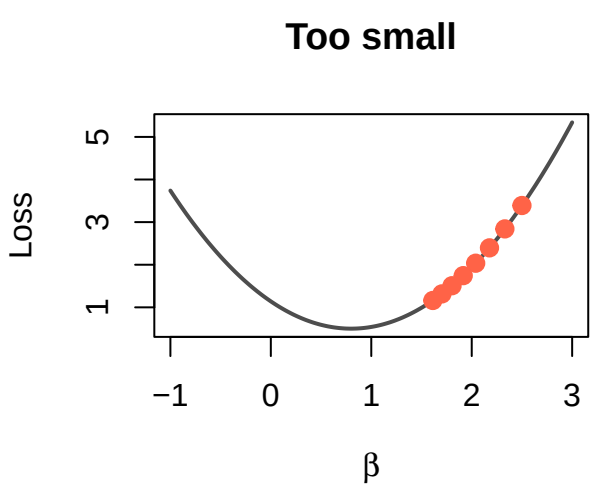
$$\beta_{t+1} = \beta_t - \alpha \cdot \nabla L(\beta_t)$$

α = **learning rate** (step size)

The Learning Rate α

Too small ($\alpha = 0.0001$)

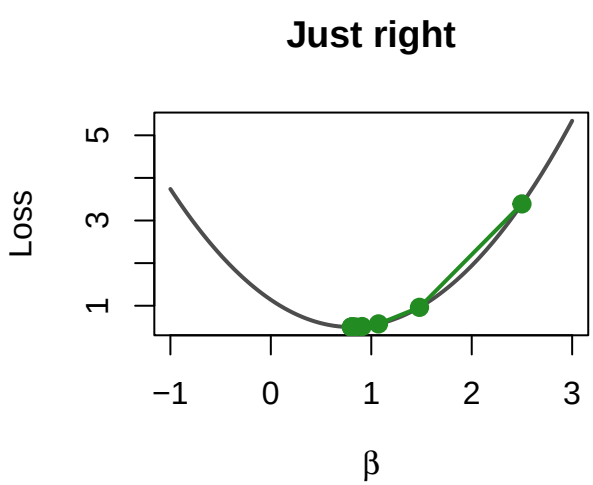
```
x_vals <- seq(-1, 3, length.out = 200)
loss_fn <- function(b) (b - 0.8)^2 + 0.5
plot(x_vals, loss_fn(x_vals), type = "l", lwd = 2, col = "gray30",
     xlab = expression(beta), ylab = "Loss", main = "Too small")
b <- 2.5; lr <- 0.05
for(i in 1:8) {
  b_new <- b - lr * 2 * (b - 0.8)
  points(b, loss_fn(b), pch = 19, col = "tomato", cex = 1.2)
  b <- b_new
}
```



Tiny steps → very slow

Just right ($\alpha = 0.003$)

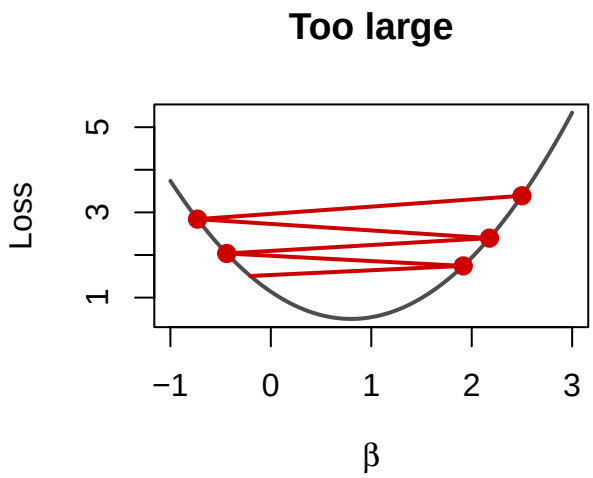
```
plot(x_vals, loss_fn(x_vals), type = "l", lwd = 2, col = "gray30",
     xlab = expression(beta), ylab = "Loss", main = "Just right")
b <- 2.5; lr <- 0.3
for(i in 1:8) {
  b_new <- b - lr * 2 * (b - 0.8)
  segments(b, loss_fn(b), b_new, loss_fn(b_new), col = "forestgreen", lwd = 2)
  points(b, loss_fn(b), pch = 19, col = "forestgreen", cex = 1.2)
  b <- b_new
}
points(b, loss_fn(b), pch = 19, col = "forestgreen", cex = 1.2)
```



Smooth convergence

Too large ($\alpha = 0.1$)

```
plot(x_vals, loss_fn(x_vals), type = "l", lwd = 2, col = "gray30",
     xlab = expression(beta), ylab = "Loss", main = "Too large")
b <- 2.5; lr <- 0.95
for(i in 1:5) {
  b_new <- b - lr * 2 * (b - 0.8)
  segments(b, loss_fn(b), b_new, loss_fn(b_new), col = "red3", lwd = 2)
  points(b, loss_fn(b), pch = 19, col = "red3", cex = 1.2)
  b <- b_new
}
```



Overshoots → diverges!

GD Trajectory: Watching β Converge

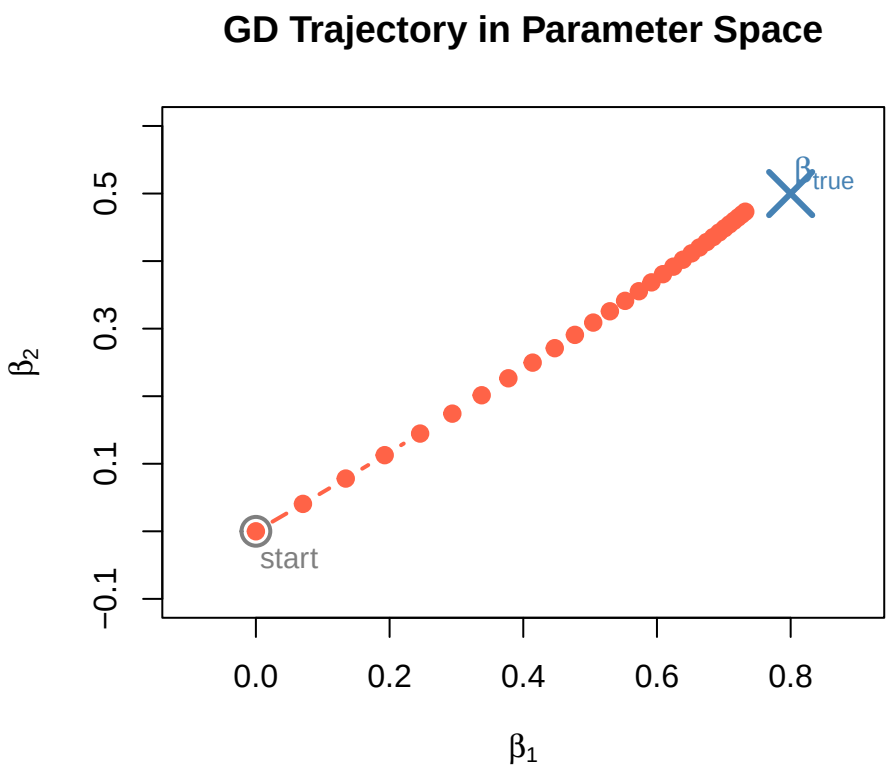
With 2 parameters (β_1, β_2), gradient descent traces a path through parameter space:

```
for _ in range(50):
    diff = x.dot(b) - y
    grad = diff.dot(x) / n
    b -= lr * grad
```

The trajectory spirals toward the true β.

```
set.seed(42)
n <- 200
x_mat <- matrix(rnorm(n * 2), n, 2)
true_coef <- c(0.8, 0.5)
y_sim <- x_mat %*% true_coef + rnorm(n, sd = 0.3)
b <- c(0, 0); lr <- 0.1
traj <- matrix(NA, 30, 2)
for(i in 1:30) {
  traj[i,] <- b
  grad <- t(x_mat) %*% (x_mat %*% b - y_sim) / n
  b <- b - lr * as.vector(grad)
}
```

```r
plot(traj[,1], traj[,2], type = "b", pch = 19, col = "tomato", lwd = 2,
     xlab = expression(beta[1]), ylab = expression(beta[2]),
     main = "GD Trajectory in Parameter Space",
     xlim = c(-0.1, 0.9), ylim = c(-0.1, 0.6))
points(true_coef[1], true_coef[2], pch = 4, col = "steelblue", cex = 3, lwd = 3)
text(true_coef[1]+0.05, true_coef[2]+0.03, expression(beta["true"]), col = "steelblue", cex =
points(0, 0, pch = 1, col = "gray50", cex = 2, lwd = 2)
text(0.05, -0.04, "start", col = "gray50", cex = 0.9)
```



Stochastic Gradient Descent (SGD)

In practice, datasets are huge — computing gradients on **all** data is expensive.

. . .

Solution: use a random **mini-batch** at each step.

$$\nabla L \approx \frac{1}{|B|} \sum_{i \in B} \nabla L_i$$

. . .

Full-batch GD:

- Exact gradient
- Slow per step
- Smooth path

SGD (mini-batch):

- Approximate gradient
- Fast per step
- Noisy but effective path

. . .

The noise actually **helps** escape bad local minima!

Three Components of Machine Learning

Every ML model needs these three ingredients:

. . .

```
 MODEL              LOSS             OPTIMIZER

ŷ = f(x)      L(ŷ, y)            ← - L

(defines        (measures how       (updates
 the             wrong we            params
 hypothesis)      are)                to
                                     reduce
                          loss)
```

. . .

Component	Linear regression	Neural network
Model	$y = X\beta$	$y = W_2\sigma(W_1x + b_1) + b_2$
Loss	MSE	MSE, cross-entropy, …
Optimizer	Normal equation	SGD, Adam, …

The Loss Function: MSE

Mean Squared Error — the most common loss for regression:

$$L(\beta) = \frac{1}{m}\sum_{i=1}^{m}(\hat{y}^{(i)} - y^{(i)})^2$$

. . .

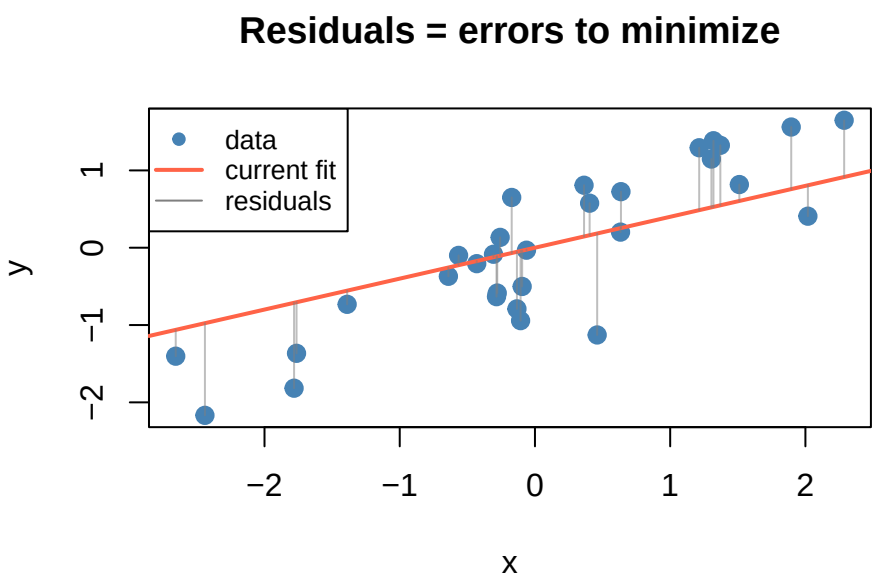
Its gradient (for linear model):

$$\frac{\partial L}{\partial \beta_j} = \frac{1}{m}\sum_{i}(\hat{y}^{(i)} - y^{(i)}) \cdot x_j^{(i)}$$

. . .

This is the "error × input" rule — the gradient is large when the error is large **and** the input contributed to it.

```
set.seed(42)
x_sim <- rnorm(30)
y_sim <- 0.8 * x_sim + rnorm(30, sd = 0.5)
y_hat <- 0.4 * x_sim  # wrong beta
plot(x_sim, y_sim, pch = 19, col = "steelblue", cex = 1.2,
     xlab = "x", ylab = "y", main = "Residuals = errors to minimize")
abline(0, 0.4, col = "tomato", lwd = 2)
segments(x_sim, y_sim, x_sim, y_hat, col = adjustcolor("gray50", 0.5), lwd = 1)
legend("topleft", c("data", "current fit", "residuals"),
       col = c("steelblue", "tomato", "gray50"),
       pch = c(19, NA, NA), lty = c(NA, 1, 1), lwd = c(NA, 2, 1), cex = 0.8)
```

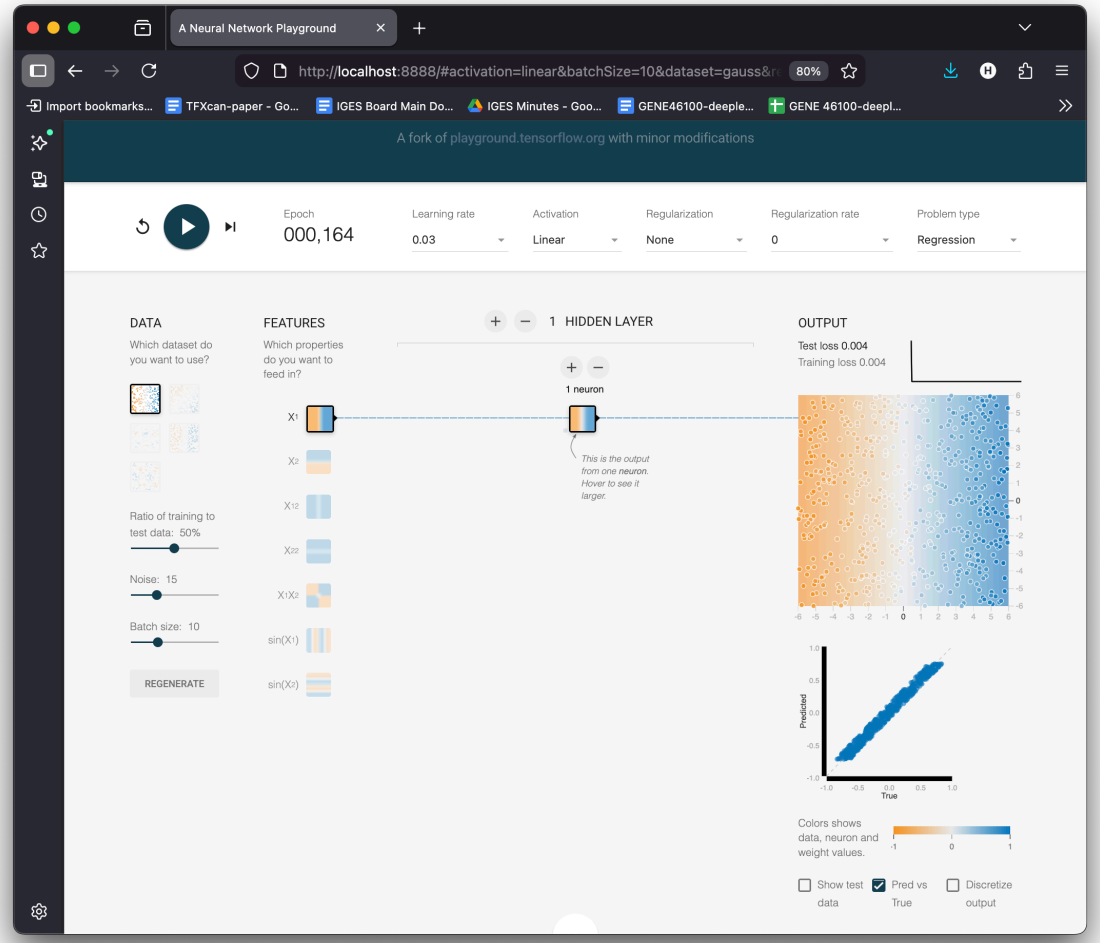


See It Live: Fit a Line in the Playground

Open **playground.hakyimlab.org**

Settings:

- Problem type → **Regression**
- Dataset → **reg-plane** (linear data)
- 1 hidden neuron, **Linear** activation
- Hit Play



. . .

Watch the **loss curve** drop — gradient descent is finding w and b right now.

Try changing the learning rate — see it converge faster or explode.

Linear Model Succeeds — On Linear Data

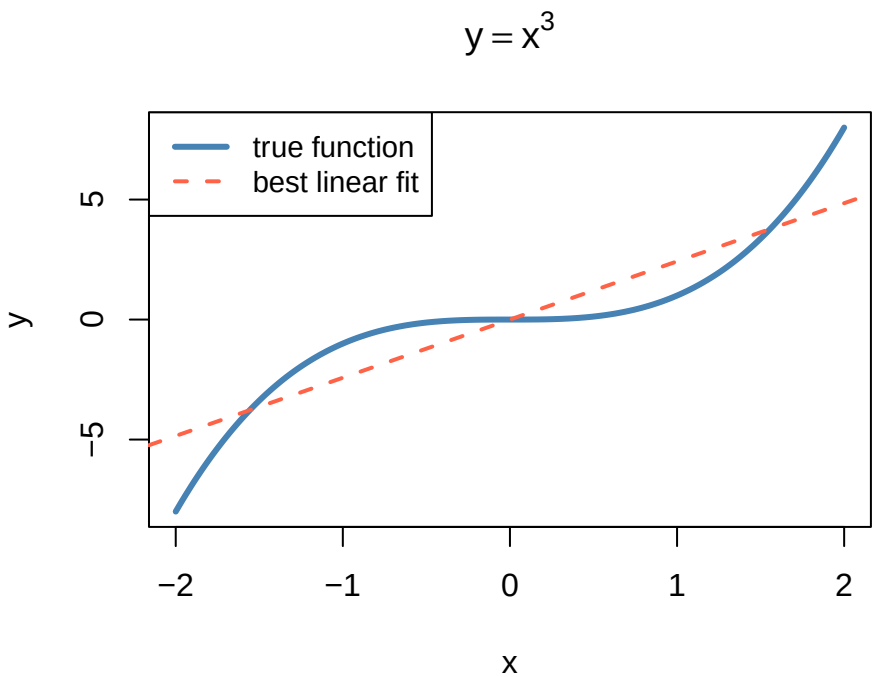
The playground fits $y = wx + b$ perfectly on the linear dataset.

. . .

But what if the data isn't linear?

. . .

```
set.seed(42)
x_nl <- seq(-2, 2, length.out = 200)
y_nl <- x_nl^3
plot(x_nl, y_nl, type = "l", lwd = 3, col = "steelblue",
     xlab = "x", ylab = "y", main = expression(y == x^3))
# best linear fit
b_fit <- coef(lm(y_nl ~ x_nl))
abline(b_fit, col = "tomato", lwd = 2, lty = 2)
legend("topleft", c("true function", "best linear fit"),
       col = c("steelblue", "tomato"), lwd = c(3, 2), lty = c(1, 2), cex = 0.9)
```



The **best possible** linear fit to $y = x^3$ is still terrible.

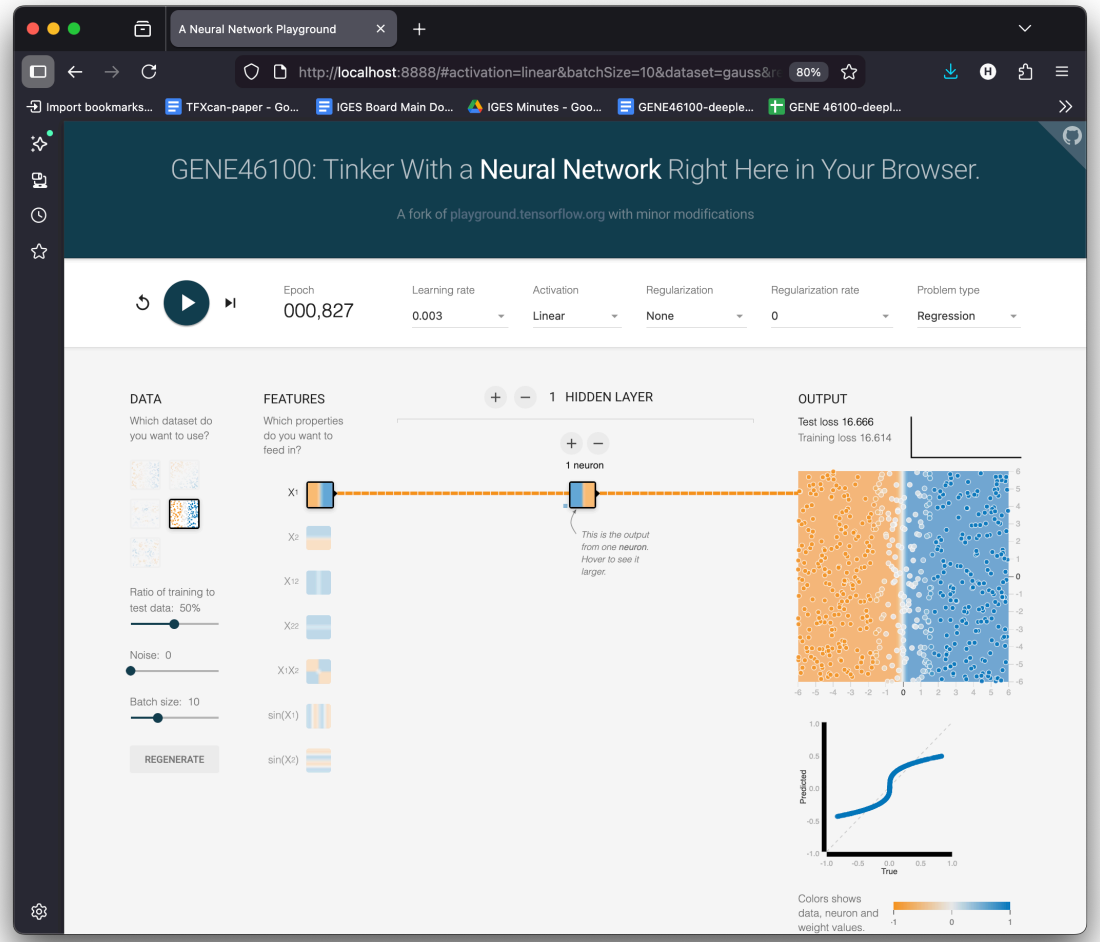
No amount of training will fix this — the **model** is wrong, not the optimizer.

Try Cubic Data in the Playground

Switch to dataset → **Cubic** ($y \approx x^3$)

Keep: 1 neuron, **Linear** activation

Hit Play and wait…



. . .

Loss stays high. The best linear fit is a flat plane through a curve.

No matter how long you train — a linear model can't bend.

Why Linear Models Fail on Nonlinear Data

The linear model can only learn:

$$y = w_1 x_1 + w_2 x_2 + b$$

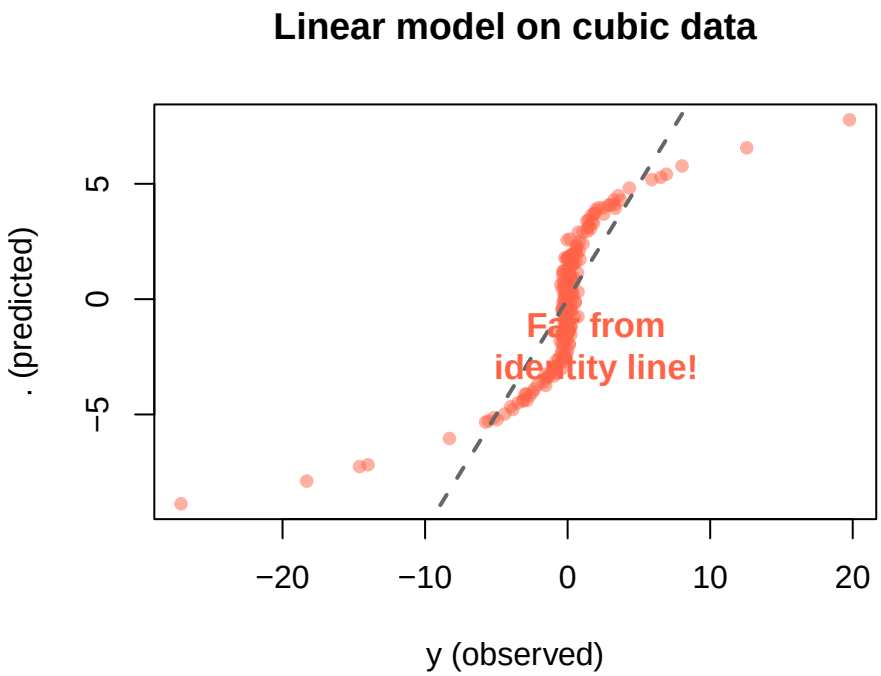
This is a **plane**. It cannot curve, twist, or bend.

. . .

What we need:

- A model that can learn *any* shape
- Without us specifying the functional form
- From data alone

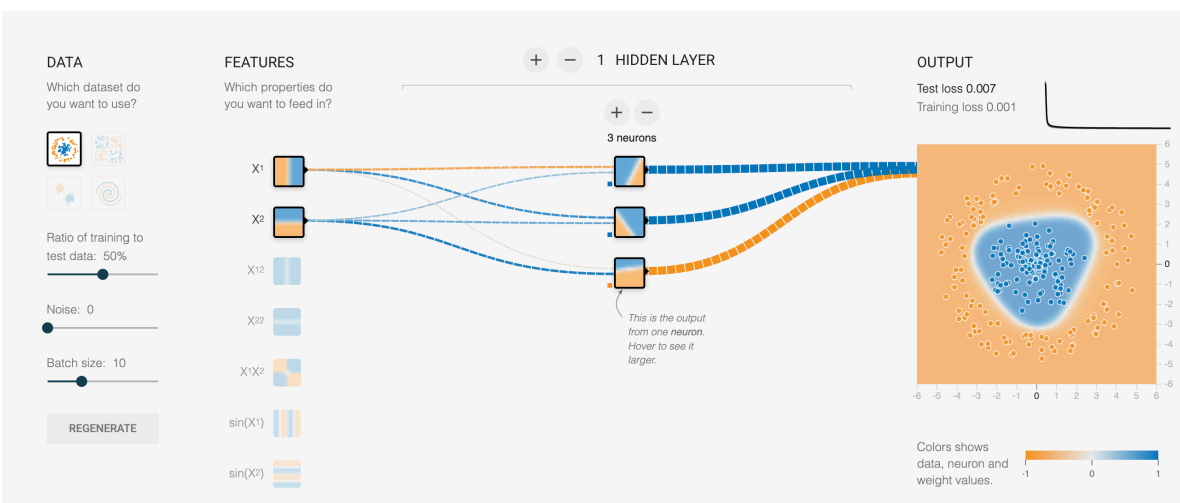
```
set.seed(42)
x_nl <- rnorm(200)
y_nl <- x_nl^3 + rnorm(200, sd = 0.3)
y_hat_lin <- coef(lm(y_nl ~ x_nl))[1] + coef(lm(y_nl ~ x_nl))[2] * x_nl
plot(y_nl, y_hat_lin, pch = 19, col = adjustcolor("tomato", 0.5), cex = 0.8,
     xlab = "y (observed)", ylab = "ŷ (predicted)",
     main = "Linear model on cubic data")
abline(0, 1, col = "gray40", lwd = 2, lty = 2)
text(2, -2, "Far from\nidentity line!", col = "tomato", cex = 1.1, font = 2)
```



. . .

Solution: add nonlinearity → build a neural network.

Multi-Layer Perceptron (MLP)



A neural network with:

- **Input layer**: your features
- **Hidden layer(s)**: with nonlinear activation
- **Output layer**: prediction

$$y = W_2 \cdot \sigma(W_1 x + b_1) + b_2$$

. . .

Key insight: σ (activation function) introduces the **bends** that let the model fit curves.

What Does a Neuron Do?

```
                        activation
  x   w            function
                 Σ     (·)    output
  x   w     +b
```

. . .

Without activation (linear):

output $= w_1 x_1 + w_2 x_2 + b$

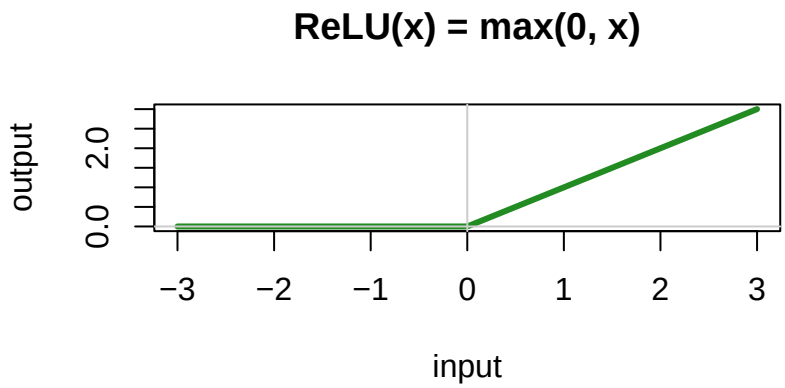
Just a weighted sum — still a line!

With ReLU activation:

output $= \max(0, w_1 x_1 + w_2 x_2 + b)$

Now it can produce a **kink** — a bent line!

```r
x_vals <- seq(-3, 3, length.out = 200)
relu <- pmax(0, x_vals)
plot(x_vals, relu, type = "l", lwd = 3, col = "forestgreen",
     xlab = "input", ylab = "output", main = "ReLU(x) = max(0, x)")
abline(h = 0, col = "gray80"); abline(v = 0, col = "gray80")
```

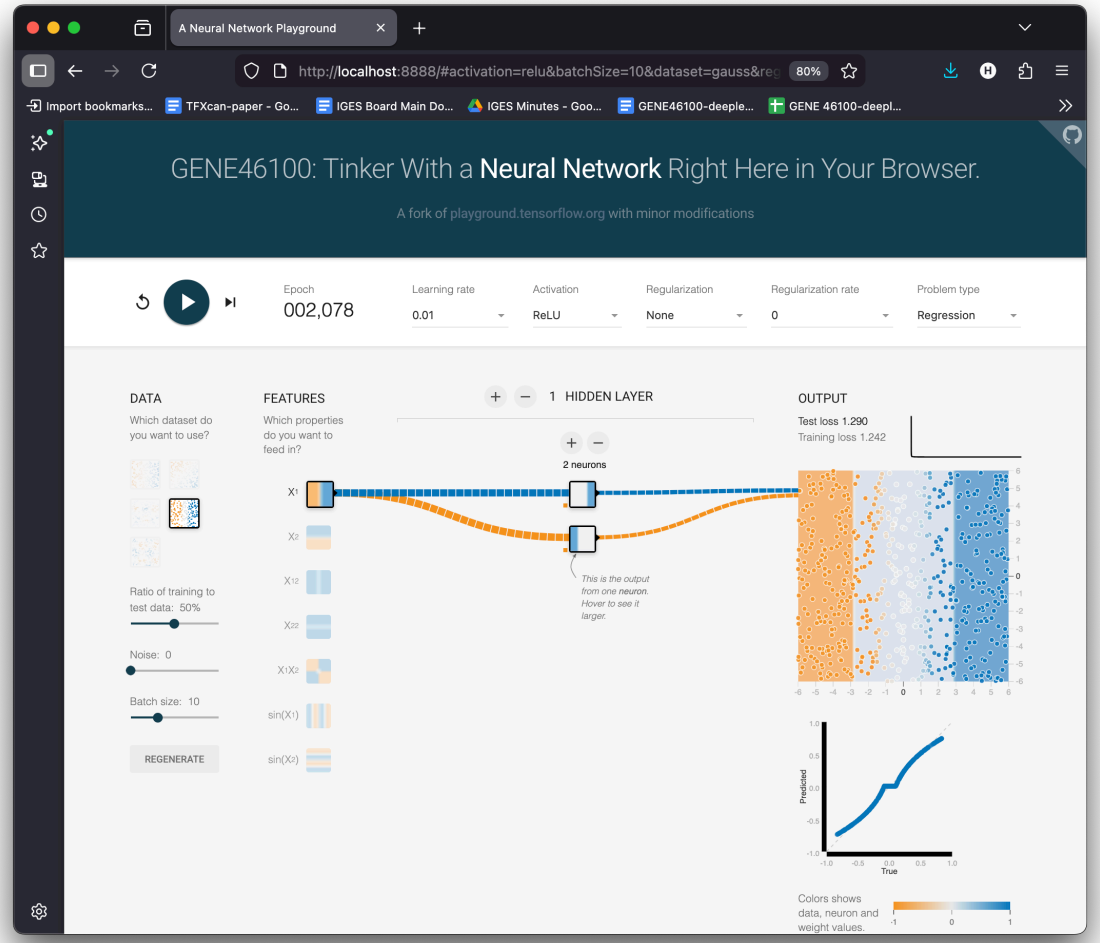


Building Up: 2 Neurons = 2 Bends

Change activation → **ReLU**

Set hidden layer → **2 neurons**

Hit Play



. . .

Each neuron contributes one ReLU "kink." Combined, they approximate a curve — but it's rough.

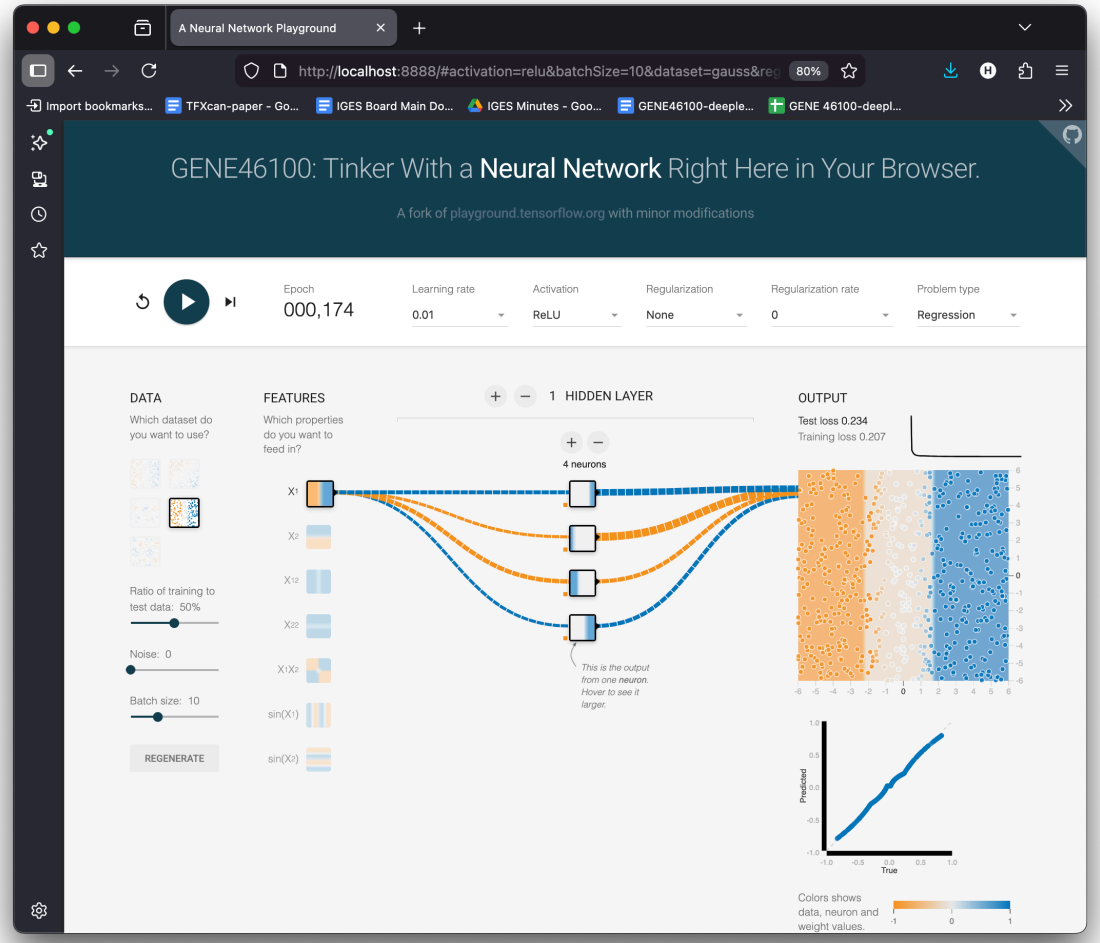
Loss is lower, but the fit isn't great yet.

---

4 Neurons = 4 Bends = Smoother

Increase to **4 neurons**

Hit Play



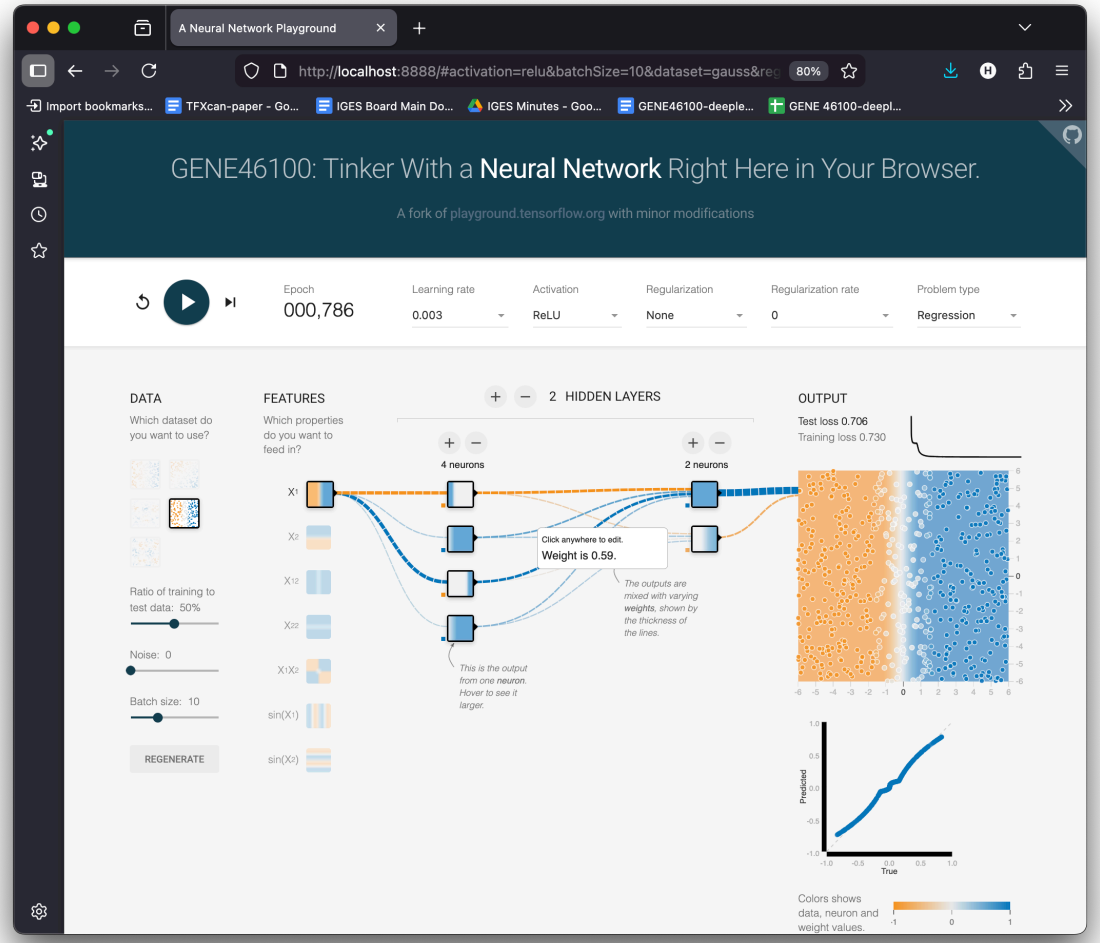
. . .

More neurons = more "kinks" = smoother approximation of x^3.

Adding Depth: 2 Layers (4 → 2)

Add a second layer: **4 neurons → 2 neurons**

Hit Play



. . .

Layer 1 learns basic bends → Layer 2 **combines** them into a smooth curve.

Stacking layers = composing simple features into complex functions.

The MLP Progression — Summary

```
LINEAR (1 neuron, no activation)          2 NEURONS + ReLU
```

```
    flat line
    can't bend!                                two bends


4 NEURONS + ReLU                           2 LAYERS (4→2) + ReLU



    four bends                                 smooth curve
```

. . .

A neural network approximates any function by combining simple bent lines.

Counting Parameters

Exercise: count parameters for this network:

- Input: 2 features
- Hidden: 3 neurons
- Output: 1

```
x      h
         h      ŷ
x      h
```

. . .

Layer 1: 2×3 weights + 3 biases = **9**

Layer 2: 3×1 weights + 1 bias = **4**

Total: 13 parameters

. . .

Manually computing gradients for 13 parameters is tedious. For millions? Impossible.

This is why we need PyTorch.

Why the Activation Function Matters

Without activation, stacking layers is pointless:

$$W_2(W_1x) = (W_2W_1)x = W'x$$

. . .

Any number of linear layers = one linear layer!

. . .

Try it yourself: set activation to "Linear" in the playground. Add as many layers as you want — the output is always a flat gradient.

The activation function **breaks** linearity, allowing each layer to add new "bends."

Common activations:

- **ReLU**: $\max(0, x)$ — fast, simple
- **Tanh**: $\frac{e^x - e^{-x}}{e^x + e^{-x}}$ — S-curve
- **Sigmoid**: $\frac{1}{1+e^{-x}}$ — for probabilities

Universal Approximation Theorem

> An MLP with a single hidden layer can approximate any continuous function to arbitrary accuracy, given enough hidden neurons.

. . .

In playground terms: with enough neurons, you have enough bends to trace any curve.

. . .

What this doesn't tell you:

- How many neurons you need
- Whether it will learn efficiently
- Whether it will generalize to new data

. . .

More: neuralnetworksanddeeplearning.com/chap4.html

From Playground to PyTorch

The playground does everything behind the scenes. In PyTorch, **you** write it:

. . .

1. Define the model:

```
class MLP(nn.Module):
  def __init__(self, input_dim, hid_dim,
               output_dim):
    super().__init__()
    self.fc1 = nn.Linear(input_dim, hid_dim)
    self.fc2 = nn.Linear(hid_dim, output_dim)

  def forward(self, x):
    x = F.relu(self.fc1(x))
    return self.fc2(x).squeeze(1)
```

2. Train it:

```
model = MLP(1, 1024, 1)
optimizer = torch.optim.SGD(
    model.parameters(), lr=0.001)
loss_fn = nn.MSELoss()

for epoch in range(10000):
    y_hat = model(x)        # forward
    loss = loss_fn(y_hat, y) # loss
    loss.backward()          # gradients
    optimizer.step()         # update
    optimizer.zero_grad()    # reset
```

PyTorch Playground

PyTorch code	Playground equivalent
`MLP(input_dim, hid_dim, output_dim)`	Network architecture (boxes)
`F.relu(...)`	Activation dropdown

PyTorch code	Playground equivalent
`model(x)`	Data flows through network
`loss_fn(y_hat, y)`	Loss number updates
`loss.backward()`	Gradients computed (invisible in playground)
`optimizer.step()`	Weights change, output updates
`lr=0.001`	Learning rate slider

. . .

PyTorch gives you full control over what the playground hides.

The Training Loop — Visually

```
Input Data

Forward Pass  ← model(x)

Compute Loss  ← loss_fn(ŷ, y)

Backward      ← loss.backward()
(Gradients)     PyTorch autograd!

Update Params ← optimizer.step()
```

Each iteration of this loop = one "tick" of the playground.

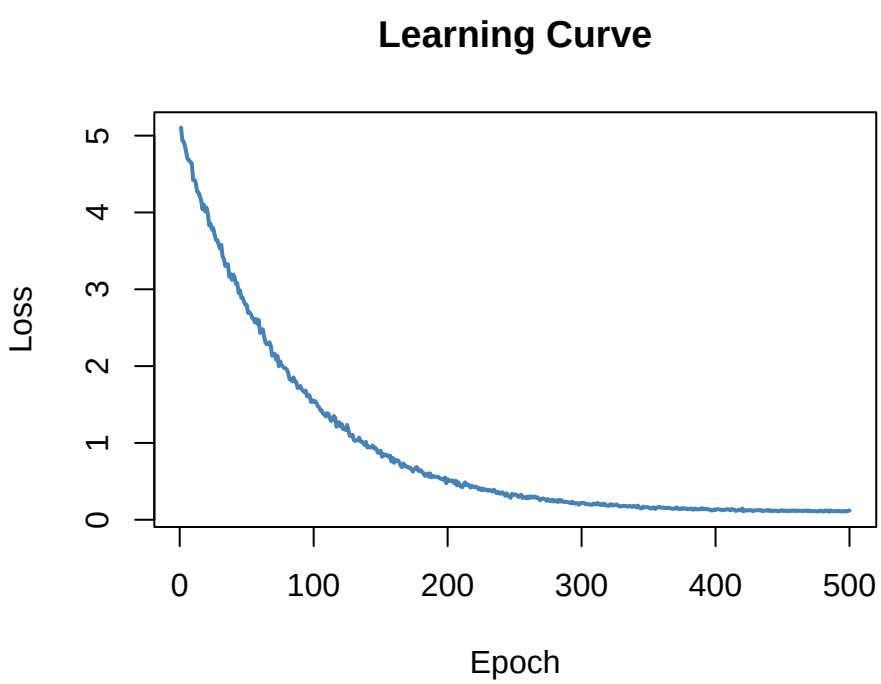
The Learning Curve

Plot **loss vs epoch** to monitor training:

```
learning_curve = []
for epoch in range(10000):
    ...
    learning_curve.append(loss.item())

qplot(range(10000), learning_curve,
      xlab="epoch", ylab="loss")
```

```
epochs <- 1:500
loss <- 5 * exp(-epochs / 80) + 0.1 + rnorm(500, sd = 0.05) * exp(-epochs/200)
loss <- pmax(loss, 0.08)
plot(epochs, loss, type = "l", col = "steelblue", lwd = 2,
     xlab = "Epoch", ylab = "Loss", main = "Learning Curve")
```



. . .

- **Dropping** = model is learning
- **Flat** = converged (or stuck)
- **Increasing** = learning rate too high!

MLP Succeeds on Cubic Data!

After training an MLP with 1024 hidden neurons on $y = x^3$:

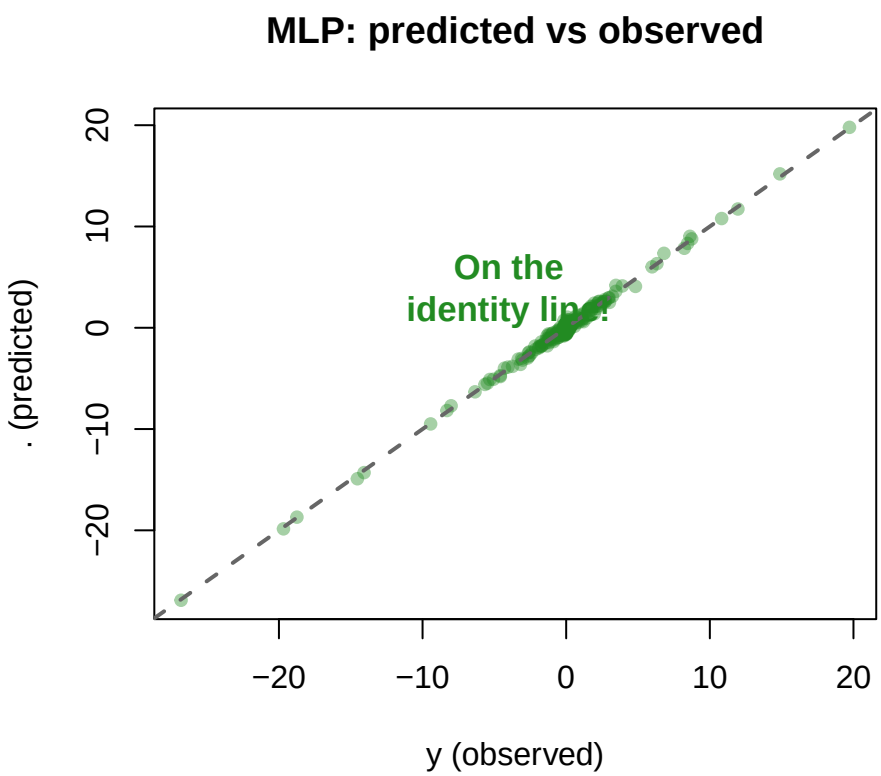
```
mlp = MLP(input_dim=1,
          hid_dim=1024,
          output_dim=1)
```

. . .

Predicted vs observed now hugs the identity line!

Compared to the linear model's failure — the MLP learned the cubic relationship from data alone.

```
set.seed(42)
x_nl <- rnorm(300)
y_nl <- x_nl^3
# Simulate a good MLP fit (add small noise)
y_hat_good <- y_nl + rnorm(300, sd = 0.3)
plot(y_nl, y_hat_good, pch = 19, col = adjustcolor("forestgreen", 0.4), cex = 0.8,
     xlab = "y (observed)", ylab = "ŷ (predicted)",
     main = "MLP: predicted vs observed")
abline(0, 1, col = "gray40", lwd = 2, lty = 2)
text(-4, 4, "On the\nidentity line!", col = "forestgreen", cex = 1.1, font = 2)
```



Your Turn: Explore the Playground

Open playground.hakyimlab.org and try:

1. **Regression → Cubic**: increase neurons from 1 to 8 — watch the fit improve
2. **Switch to Classification → Circle**: same idea, different task
3. **Set activation to Linear** — can it ever fit a curve or separate a circle?
4. **Crank learning rate to 1.0** — what happens to the loss?
5. **Turn on "Show test data"** — is a big network overfitting?

DNA as Data: One-Hot Encoding

How do we feed DNA to a neural network?

```
Sequence:  A  T  G  C  G  T  A

           A  T  G  C
     1:  [ 1  0  0  0 ]  ← A
     2:  [ 0  1  0  0 ]  ← T
     3:  [ 0  0  1  0 ]  ← G
     4:  [ 0  0  0  1 ]  ← C
     5:  [ 0  0  1  0 ]  ← G
     6:  [ 0  1  0  0 ]  ← T
     7:  [ 1  0  0  0 ]  ← A
```

Shape: (sequence_length, 4)

Each base → a 4-dimensional unit vector. No ordinal relationship imposed.

From DNA to Prediction

```
DNA sequence          One-hot matrix       Neural network      Prediction

ATGCGTAACG...  →             →             →   binding score
                    1 0 0 0               MLP or          expression level
                    0 1 0 0                CNN            variant effect
                    0 0 1 0                                ...
                    0 0 0 1
                    ...
```

. . .

This is the core pattern of the entire course.

Every model we study takes DNA sequence as input and predicts a biological output.

MLP vs CNN: A Preview

MLP (this unit)

- Flatten DNA → single vector
- Every input connected to every neuron
- No concept of "position"
- Good for learning the basics

CNN (next)

- Preserve sequential structure
- Sliding window = **motif scanner**
- Learned filters PWMs
- The workhorse of genomic DL

. . .

A CNN filter sliding over one-hot DNA is mathematically equivalent to scoring with a **Position Weight Matrix (PWM)** — but learned from data!

Unit 0 Plan: Weeks 1–3

Session	Topic	Notebook
Week 1a	Setup, intro	*(this lecture)*
Week 1b	Linear → MLP in PyTorch	`hands-on-introduction_to_deep_learning.ipyn`
Week 2	CNN for DNA scoring	`updated-basic_DNA_tutorial.ipynb`
Week 3a	TF binding project	`tf-binding-prediction-starter.ipynb`
Week 3b	Hyperparameter tuning	`tf-binding-wandb.ipynb`

What's Coming Next

Unit 1: Transformers & GPT

- Attention mechanism
- Karpathy's nanoGPT
- Train a DNA language model
- Fine-tune for promoter prediction

Unit 2: Enformer & Borzoi

- Predict epigenome from 200kb DNA
- Variant effect prediction
- Connection to GWAS/PrediXcan

. . .

The arc: MLP → CNN → Transformer → Genomic foundation models

Resources

Videos:

- 3Blue1Brown: Neural Networks — beautiful visual intuition
- Karpathy: Building GPT from scratch — we'll use this in Unit 1

Interactive:

- playground.hakyimlab.org — experiment with MLP architecture interactively
- playground-guide.md — reference for all playground controls

Papers (optional):

- Avsec et al. 2021 — Enformer
- Linder et al. 2023 — Borzoi

Getting Started

Environment: Google Colab (GPU provided, no local setup needed)

First notebook: `hands-on-introduction_to_deep_learning.ipynb`

```
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
```

Questions?